

COSC 594

DM for Robotic Planning (III)

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Outline

- RECAP
- MULTI-ROBOT MOTION PLANNING
- DIFFUSION MODEL FOR MULTI-ROBOT MOTION PLANNING

Outline

■ RECAP

Conditional diffusion model

Unconditional diffusion model: $p_{\theta}(\tau^{t-1} | \tau^t)$

Conditional diffusion model: $p_{\theta}(\tau^{t-1} | \tau^t, h)$ w/ task info h

■ BACKWARD GENERATIVE PROCESS

$$p_{\theta}(\tau^{0:T} | h) = p(\tau^T | h) \prod_{t=1}^T p_{\theta}(\tau^{t-1} | \tau^t, h)$$

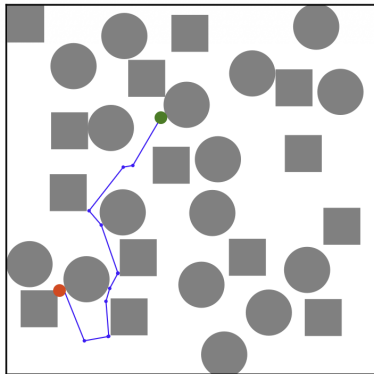
$p(\tau^T | h)$ – standard Gaussian

★ task-conditioned posterior

$$p_{\theta}(\tau^{t-1} | \tau^t, h) \propto \underbrace{p(h | \tau^t, \tau^{t-1})}_{\text{task likelihood } p(h | \tau^{t-1})} \times \underbrace{p_{\theta}(\tau^{t-1} | \tau^t)}_{\text{unconditional model/prior}}$$

Single-robot navigation

■ POINT-MASS2D DENSE



Random start/goal positions

Diffusion model composition for planning

K diffusion models: $\mu_{\theta^k}(\tau^t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(\tau^0)$ for $k = 1, \dots, K$

Reward model: $r(\tau^0)$ for a sample τ^0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{\tau^0 \sim p(\cdot)}[r(\tau^0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot))$$

$\beta_1, \dots, \beta_K$ – tradeoff parameters

★ optimal model $p^\star$

$$p^\star(\tau^0) \propto p^{\text{con}}(\tau^0) e^{r(\tau^0)/\mathbf{1}^\top \beta} \quad \text{and} \quad p^{\text{con}}(\tau^0) = \prod_{k=1}^K (p^k(\tau^0))^{\beta_k / \mathbf{1}^\top \beta} \times \frac{1}{Z}$$

Exponentially weighted conjunction distribution

Diffusion model composition as inference

K diffusion models: $\mu_{\theta^k}(\tau^t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(\tau^0)$ for $k = 1, \dots, K$

Reward model: $r(\tau^0)$ for a sample τ^0

■ BAYESIAN PERSPECTIVE

$$p^*(\tau^0) \propto \underbrace{p^{\text{con}}(\tau^0) e^{r(\tau^0)/\mathbf{1}^\top \beta}}_{\text{evidence}} \quad \text{and} \quad \underbrace{p^{\text{con}}(\tau^0)}_{\text{prior}} = \prod_{k=1}^K \left(p^k(\tau^0) \right)^{\beta_k / \mathbf{1}^\top \beta} \times \frac{1}{Z}$$

Optimal model p^* is **a posterior distribution**

Outline

- MOTIVATION

- MULTI-ROBOT MOTION PLANNING

Trajectory planning

System dynamics: $s_{t+1}^i = f^i(s_t^i, a_t^i)$ for $i = 1, \dots, m$



i -th trajectory $\tau^i = (s_0^i, a_0^i, s_1^i, a_1^i, \dots, s_T^i, a_T^i)$

■ REQUIREMENTS

- ★ energy efficiency
- ★ obstacle avoidance
- ★ trajectory smoothness

Trajectory optimization

■ ACTION OPTIMIZATION

$$\underset{\tau}{\text{maximize}} \quad R(\tau; \mathbf{s}_0) \quad := \quad \sum_{i=1}^m \sum_{t=0}^T r^i(s_t^i, a_t^i)$$

reward, e.g., negative total costs of collision

- ★ an optimal action sequence $\mathbf{a}_{0:T}^{\star} := (a_{0:T}^{1,\star}, \dots, a_{0:T}^{m,\star})$
- ★ generate an optimal trajectory $\tau^{\star} := (\tau^{1,\star}, \dots, \tau^{m,\star})$

$$\tau^{i,\star} = (\mathbf{s}_0^i, a_0^{i,\star}, s_1^i, a_1^{i,\star}, \dots, s_T^i, a_T^{i,\star})$$

- ★ task info $\mathbf{h} := (h^1, \dots, h^m)$, e.g., $(s_0^1, \dots, s_0^m), (s_T^1, \dots, s_T^m)$

Planning as inference

Random trajectories: $\tau^i = (s_0^i, a_0^i, s_1^i, a_1^i, \dots, s_T^i, a_T^i)$ for $i = 1, \dots, m$

Posterior trajectory distribution: $p(\tau^i | h) \propto p(h | \tau^i) p(\tau^i)$

$p(h | \tau^i)$ – likelihood of completing tasks

$p(\tau^i)$ – prior

■ MAXIMUM A POSTERIORI

$$\underset{\tau^i}{\text{maximize}} \quad p(\tau^i | h)$$



$$\underset{\tau^i}{\text{maximize}} \quad \log p(\tau^i | h)$$

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Sampling from conditional diffusion model

Unconditional diffusion model: $p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t})$

Conditional diffusion model: $p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t}, h)$ w/ task info h

■ UNCONDITIONAL MODEL/PRIOR

$$\log p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t}) = \log \mathcal{N}(\tau^{i,t-1}; \mu_{\theta}^i(\tau^{i,t}, t), \sigma^2(t)I) \approx -\frac{1}{2} \|\tau^{i,t-1} - \mu_{\theta}^i(\tau^{i,t}, t)\|^2$$

■ TASK LIKELIHOOD

$$\log p^i(h | \tau^{i,t-1}) \approx \log p^i(h | \mu_{\theta}^i(\tau^{i,t}, t)) + \underbrace{(\tau^{i,t-1} - \mu_{\theta}^i(\tau^{i,t}, t)) \nabla_{\tau} \log p(h | \mu_{\theta}^i(\tau^{i,t}, t))}_{\text{guidance } g^i}$$

■ TASK-CONDITIONED POSTERIOR

$$\begin{aligned}\log p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t}, h) &\approx \log p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t}) + \log p^i(h | \tau^{i,t}, \tau^{i,t-1}) \\ &= -\frac{1}{2} \|\tau^{i,t-1} - \mu_{\theta}^i(\tau^{i,t}, t)\|^2 + (\tau^{i,t-1} - \mu_{\theta}^i(\tau^{i,t}, t)) g^i \\ &\approx -\frac{1}{2} \|\tau^{i,t-1} - \mu_{\theta}^i(\tau^{i,t}, t) - g^i\|^2\end{aligned}$$

$\Downarrow$

$$p_{\theta}^i(\tau^{i,t-1} | \tau^{i,t}, h) = \mathcal{N}(\tau^{i,t-1}; \mu_{\theta}^i(\tau^{i,t}, t) + g^i, \bar{\sigma}^2(t)I)$$

Guidance g^i captures info: initial/final states, smoothness, obstacles, etc

Imposing soft-constraints as guidance

Guidance: $g^i = \nabla_{\tau} \log p^i(h \mid \tau^{i,t-1})|_{\tau^{i,t-1}=\mu_{\theta}(\tau^{i,t},t)}$

■ INITIAL/FINAL STATES

$$p^i(h \mid \tau^i) = e^{-\|s_0^i - \tau^i[0]\|^2} \implies \nabla_{\tau} \log p^i(h \mid \tau^{i,t-1}) = -\|s_0^i - \tau^i[0]\|^2$$

$$p^i(h \mid \tau^i) = e^{-\|s_T^i - \tau^i[T]\|^2} \implies \nabla_{\tau} \log p(h^i \mid \tau^{i,t-1}) = -\|s_T^i - \tau^i[T]\|^2$$

■ COLLISION COST

$$p^i(h \mid \tau^i) = e^{-c^i(\tau^i)} \implies \nabla_{\tau} \log p^i(h \mid \tau^{i,t-1}) = -c^i(\tau^i)$$

Differentiable over trajectory $\implies$ **tractable to compute** $g^i = \nabla_{\tau} \log p^i(h \mid \tau^i)$

When and where impose constraints?

■ SEQUENTIAL PLANNING

$$\tau^1 \rightarrow \tau^2 \rightarrow \tau^3 \rightarrow \dots \rightarrow \tau^m$$

■ SEQUENTIAL PLANNING

$(\tau_t^1, \tau_t^2, \dots, \tau_t^m) \rightarrow$ find a collision time t and impose constraint

Experiment I: Highways map

■ MULTI-ROBOT DATA

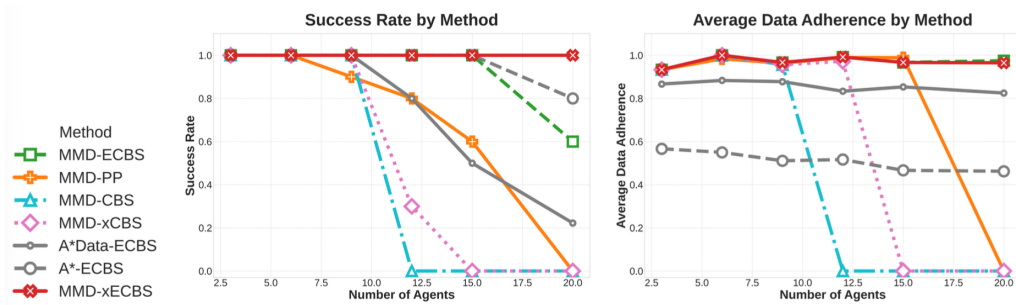


Counter-clockwise movement around a central obstacle

image source: arXiv: 2410.03072

Experiment I: Highways map

■ SUCCESS RATES AND DATA ADHERENCE



Multi-robot multi-model planning diffusion (MMD-xECBS) outperforms

Experiment II: A larger environment

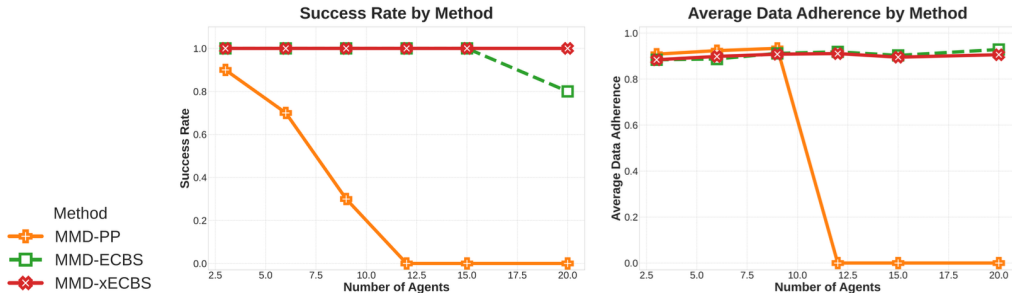
■ LARGER MAP



Expert data trajectories

Experiment II: A larger environment

■ DIFFUSION TRAJECTORIES



Multi-robot multi-model planning diffusion (MMD-xECBS) scales

Summary

- RECAP

- MULTI-ROBOT MOTION PLANNING

 - Trajectory optimization, planning as inference

- DIFFUSION MODEL FOR MULTI-ROBOT MOTION PLANNING

 - Conditional diffusion model, imposing soft-constraints